

The Framingham Heart Study: Risk Factors for and Probability of Stroke by Sex

Introduction

The Framingham Heart Study, which began in 1948, was a first of its kind long-term prospective study of the etiology of cardiovascular disease among a population of free-living subjects in the community of Framingham, Massachusetts. A subset of the data collected on this study, during three examination periods, approximately 6-years apart, is analyzed here.

The main objective of this analysis is to test for risk factors associated with developing stroke from a list of demographic, health indicators, and clinical marker variables—all of which were measured at baseline; and then estimate 10-year probabilities of having stroke for different risk profiles. This analysis will be performed separately for males and females. Secondly, since participants were examined at 3 time periods, it is of interest to describe how diabetes status and systolic blood pressure (BP) changes across these time periods.

Methods

Mean and standard deviation estimates were reported for continuous variables while frequency and proportion were reported for categorical variables. These estimates were tabulated and stratified by sex; a column showing the marginal distribution of the variables was also included in the table. Inference was performed to compare the distribution of continuous and categorical variables by sex using appropriate statistical methods. In a separate table, descriptive measures were obtained by combination of sex and stroke event status as the stratification/column variable.

The primary endpoint is time to stroke; it is defined as time from study entry to the time of documentation of a stroke event during follow-up. Follow-up time in days was scaled to years by dividing by 365.25. Study participants that do not develop stroke (or died) were censored at the last study assessment period (or death). Participants who had prevalent stroke at baseline were

excluded from the analysis cohort. Analysis was limited to the first 10 years of follow-up

because risk factors were measured more closely in time to this period. For this reason, follow-up time beyond 10 years was censored at 10 years. Likewise, stroke events that occurred beyond 10-years were censored at 10 years. For participants without a stroke event and with documented stroke time longer than death time, stroke time was recoded to be the same as death time.

Among the variables available in the dataset, the subset that comprises the variables of interest (*VoI*) include age, diabetes status, systolic blood pressure (mmHg), use of anti-hypertensive meds, prevalent coronary heart disease (CHD), current cigarette smoking status, serum total cholesterol (mg/dL), and body mass index (kg/m²). Different participant analysis populations were created: **(a)** the clean data comprises all participants without prevalent stroke at baseline, **(b)** the model/variable selection data comprises all participants in the clean data, from (a), and without missing values for any of the 8 *VoI*, and **(c)** the final data, also created from the clean data in (a) comprised all participants without missing data for the subset *VoI* deemed associated with the outcome (methods detailed next). A common dataset, as defined in (b), was used for variable selection for consistency in the estimation and comparison of model performance metrics.

For univariable analysis, Kaplan-Meier (KM) method was used to visualize the distribution of the outcome (time to stroke) by categories of a predictor—continuous predictors were grouped using common/known cutoffs or based on the pooled (men and women) median value. The KM plot includes estimated hazard ratio from the univariable Cox proportional hazards (PH) model, p-value from the Schoenfeld residual test of PH assumption, Concordance-index (C-index), and the logrank test statistic comparing time to stroke by levels of the predictor. Univariable analysis was done separately for men and women. Also, the hazard for stroke was plotted by sex using a

global bandwidth and with boundary correction at both (start and end of follow-up) tails.

Martingale residual analysis was used to assess the functional form of continuous variables¹. A two-step model building process was used to identify predictors associated with time to stroke among the 8 *VoI*. In the first step, a backwards model selection, based on the change in the Akaike Information Criterion (AIC), was used to identify variables associated with the outcome using a multivariable Cox PH model. A backwards selection approach was used because it has optimal variable selection properties compared to other common/routine model selection methods. The first 3 *VoI* predictors are important and included in the model regardless of the information criteria. In a subsequent step, and using the variables selected by the backwards approach, pairwise interactions between each of the 3 important predictors and the other 5 predictors were assessed, with a multivariable Cox PH model, using a likelihood ratio test (LRT) with a 5% significance threshold. Unless there is a strong indication of a threshold effect, continuous variables were analyzed in the model as such (continuous) except for high systolic BP which was categorized using the threshold of ≥ 160 mmHg. The model building process was done separately for men and women using the model selection data as described in (b). After the model for each sex was finalized, the identified variables were fitted using the final data described in (c) for subsequent analysis including estimation of stroke probabilities. Using the Cox PH model fit object, 10-year stroke probabilities were estimated for different participants risk profiles of research interest.

Analysis for the secondary objective of assessing change in diabetes status and systolic BP values was descriptive in nature. This analysis was limited to the participants with follow-up data across all 3 timepoints. For diabetes, transition probabilities between Period 1 (baseline) and Period 2, and between Period 2 and Period 3 was estimated. The same was done for categorized

high systolic BP, and in addition to this, change in BP values between the 2 transition periods was plotted against the respective reference period. All analysis was performed in R statistical software, using `gtsummary`, `survival`, `muhaz`, `MASS`, `gt`, and `flextable` packages.

Results

The provided data has 11627 observations among 4434 unique participants. A total of 32 unique participants making up 60 observations had prevalent stroke at baseline. After this exclusion, there were 4402 unique participants remaining. For 7 unique participants, the time to stroke was longer than the time to death, the time to stroke was censored at the date of death for these cases. After restricting follow-up, there were 111 (2.5%) stroke events; 2.5% each among men and women (logrank p-value = 0.835). The variable selection data (b) comprised 4275 unique participants: 1897 and 2378 men and women respectively. The distribution of participants characteristics is shown in **Table 1**. At baseline, age, diabetes status, and average systolic BP is comparable by sex. However, more women had high systolic BP (≥ 160 mmHg), reported more use of anti-hypertensive medication, lower prevalent CHD (**Table 1**). Also, missing data is more common among women (4%) than men (2%). For continuous variables, the mean and median estimates are similar suggesting minimal skewness. **Supplementary Table 1** shows the baseline characteristics stratified by sex and stroke event status. **Figure 1** shows Kaplan-Meier (KM) plot of time to stroke [left panel], and the baseline hazard for stroke [right panel] by sex. From the hazard plot, the hazard rate at time 0 is higher for women than men. However, the hazard for men appear to trend upward as time went on. **Supplementary Figure 1a** and **1b** shows results of univariable analysis for the 8 *VoI* for men and women, side-by-side. Except for prevalent CHD, serum total cholesterol and BMI categories, the pattern of association between the outcome and each of the variables is consistent by sex. Based on p-values, the Schoenfeld residual test of PH

does not suggest a major violation of the PH assumption. **Supplementary Figure 2** shows results of functional form assessments for the continuous *Vol* by sex. There appears to be threshold effect for average systolic BP around 150 mmHg. Note that this variable is being analyzed as a binary variable using a cutoff not too far (≥ 160 mmHg) from the indicated threshold. For the other variables, a quadratic fit was assessed which was not better than the simpler linear fit.

Among men, the backwards selection method selected age, diabetes, high systolic BP, and current smoker status. For females, it selected age, high systolic BP, and use of anti-hypertensive medications. **Supplementary Table 2** shows results for the interaction testing step, and the final statistically significant interaction models (highlighted in green) for men and women.

The final data (see definition (c) above) for men and women comprised 1908 and 2390 unique participants respectively. **Table 2** shows hazard ratio estimates from the fitted multivariable models for men and women. The final model for men includes age, diabetes status, high systolic BP, use of anti-hypertensive meds, and the interaction between high systolic BP and use of anti-hypertensive meds. Having any of these risk factors increases the risk of having stroke, but use of anti-hypertensive meds among those with high systolic BP reduces the risk of having stroke. Holding other factors constant, participants on meds for high systolic BP have 0.08 reduced odds of having stroke. The final model for females shares the same first 4 variables listed above, except with the inclusion of serum total cholesterol and an interaction between age and serum total cholesterol. As is the case among men, having any of these risk factors increases the risk of having stroke, but older women with a high serum total cholesterol have a lower risk of stroke. In both models, the interaction terms were statistically significant. Cox PH model estimates of the 10-year probability of stroke are shown in **Table 3**. Risk is lowest for the 40-year-olds in the

reference profile for men and women. Among men, it is highest among those with diabetes and high systolic BP. Among women, it is highest among those that use meds and are in the 90th percentile of serum total cholesterol. For men and women, the Schoenfeld test for the final model shows no violation of the PH assumption for any of the variables in the model.

Transition probabilities are shown in **Table 4** for diabetes and high systolic BP status. Among the subset with data throughout follow-up, those with diabetes at baseline had diabetes at subsequent follow-up timepoint, for men and women. Between 30% to 43% of those with high systolic BP transitioned to low/normal systolic BP at subsequent timepoints. Compared to baseline, the prevalence of diabetes and high systolic BP increased from 3% and 12% to between 7.2% to 8.7% and 17.3% to 22.0% respectively. **Figure 2** is a scatter plot of change in systolic BP versus a reference timepoint for men and women. Across the 4 panels, the lower left grid which indicates those with a lower systolic BP transitioned to higher BP at a subsequent timepoint, and vice versa for those with a high BP.

Discussion

Several risk factors associated with developing stroke were identified among men and women. The model in the respective groups has good discrimination statistics, C-index = 0.795. Besides the 3 face-value variables that were required to be included in the model, the use of anti-hypertensive medications was included in the models for men and women. The hazard ratio for the relationship between this risk factor and outcome is much higher in magnitude among men compared to women. However, this may be due to small sample size since the prevalence of medication use and high systolic BP is lower in men than women (**Table 1** and **Supp. Table 1**). Serum total cholesterol was included in the model for women, but not for men. The reason for this may be the large magnitude of difference in mean serum cholesterol among women with

stroke event vs those without compared to the difference in men (**Supp. Table 1**). Although diabetes status was not a statistically significant predictor of time to stroke among women which may be due to low stroke events and therefore low statistical power, the magnitude of its effect is large and its prevalence changed from 3% at baseline to 7.8% at the 2nd follow-up. The same is true for high systolic BP. For these reasons, it can be concluded that there is substantial change in patients' status, therefore a time-varying analysis is warranted.

There are several limitations in the current analysis. Firstly, the proportion of stroke events is low, 2.5% each among men and women. In survival analysis, this limits the complexity/number of variables that may be included in the model since failures/events are the currency of failure-time models. This fact impacted on the approach used to develop the final model presented in this report. Also, a substantial number of patients did not have follow-up throughout the 3 follow-up periods. However, this may be due to the restriction of follow-up to 10 years.

Table 1: Participants characteristics at baseline, stratified by sex

Participants' Baseline Characteristics	Participants' Sex		p-value ²	Overall N = 4,402 ¹
	Men N = 1,930 ¹	Women N = 2,472 ¹		
Age at exam (years)	49.7 (8.7), [49.0]	50.0 (8.6), [49.0]	0.271	49.9 (8.7), [49.0]
Diabetic (casual glucose $\geq$ 200 mg/dL)			0.201	
Diabetic	59 (3%)	60 (2%)		119 (3%)
Not Diabetic	1,871 (97%)	2,412 (98%)		4,283 (97%)
Average systolic blood pressure (mmHg)	131.6 (19.4), [128.5]	133.7 (24.3), [128.5]	0.438	132.8 (22.3), [128.5]
Systolic blood pressure status			<0.001	
High Systolic BP ($\geq$ 160 mmHg)	171 (9%)	353 (14%)		524 (12%)
Normal Systolic BP (<160 mmHg)	1,759 (91%)	2,119 (86%)		3,878 (88%)
Use of anti-hypertensive medication			<0.001	
Current Use	40 (2%)	96 (4%)		136 (3%)
Not Currently Used	1,868 (98%)	2,338 (96%)		4,206 (97%)
NA (missing)	22 (1.1%)	38 (1.5%)		60 (1.4%)
Prevalent Coronary Heart Disease			<0.001	
Prevalent Disease	120 (6%)	67 (3%)		187 (4%)
Free of Disease	1,810 (94%)	2,405 (97%)		4,215 (96%)
Current cigarette smoking			<0.001	
Current Smoker	1,168 (61%)	1,001 (40%)		2,169 (49%)
Not Current Smoker	762 (39%)	1,471 (60%)		2,233 (51%)
Serum total cholesterol (mg/dL)	233.6 (42.4), [231.0]	239.7 (46.2), [237.0]	<0.001	237.0 (44.6), [234.0]
NA (missing)	7 (0.4%)	45 (1.8%)		52 (1.2%)
Body Mass Index (kg/m ²)	26.2 (3.4), [26.1]	25.6 (4.5), [24.8]	<0.001	25.8 (4.1), [25.5]
NA (missing)	4 (0.2%)	13 (0.5%)		17 (0.4%)
BMI Categories			<0.001	
Underweight	12 (1%)	45 (2%)		57 (1%)
Healthy Weight	697 (36%)	1,226 (50%)		1,923 (44%)
Overweight	987 (51%)	847 (34%)		1,834 (42%)
Obese	230 (12%)	341 (14%)		571 (13%)
NA (missing)	4 (0.2%)	13 (0.5%)		17 (0.4%)
Stroke event status			0.948	
Stroke Event	49 (3%)	62 (3%)		111 (3%)
No Stroke Event	1,881 (97%)	2,410 (97%)		4,291 (97%)
Nos. missing values for key variables			<0.001	
No missing data	1,897 (98%)	2,378 (96%)		4,275 (97%)
Missing for 1 variable	33 (2%)	92 (4%)		125 (3%)
Missing for 2 variables	0 (0%)	2 (0%)		2 (0%)

¹Mean (SD), [Median]; n (%)²Wilcoxon rank sum test; Pearson's Chi-squared test; Fisher's exact test

Table 2: Estimated hazard ratio for the relationship between risk factors and stroke

Characteristic	Men			Women		
	HR	95% CI	p-value	HR	95% CI	p-value
Age (years) [unit change]	1.07	(1.03, 1.11)	<0.001	1.34	(1.105, 1.63)	0.003
Diabetic (Yes vs No)	3.81	(1.67, 8.68)	0.001	1.98	(0.71, 5.49)	0.191
High Systolic BP (≥ 160 vs < 160 mmHg)	5.56	(2.95, 10.49)	<0.001	2.4	(1.38, 4.33)	0.002
Use of Anti-hypertensive Meds. (Yes vs No)	11.26	(2.67, 47.52)	<0.001	2.05	(0.97, 4.33)	0.061
High Systolic BP & Anti-hypertensive Meds	0.08	(0.01, 0.60)	0.015	-	-	-
Serum Total Cholesterol (mg/dL)	-	-	-	1.05	(1.00, 1.10)	0.036
Age * Serum Total Cholesterol (mg/dL)	-	-	-	0.999	(0.998, 1.00)	0.031
Concordance index [SE]	0.795 [0.034]			0.795 [0.031]		

Abbreviations: CI = Confidence Interval, HR = Hazard Ratio

Table 3: Estimated 10-year stroke probabilities for different risk profiles by sex

Sex	Risk profile	Age 40	Age 50	Age 60
Male	Reference (no risk)	0.7% (0.4, 1.4)	1.4% (1.0, 2.2)	2.8% (1.8, 4.3)
“	Diabetes only	2.8% (1.0, 7.4)	5.4% (2.3, 12.1)	10.3% (4.6, 22.0)
“	High BP only	4.0% (1.8, 8.8)	7.7% (4.4, 13.5)	14.7% (8.8, 23.8)
“	Diabetes & High BP	14.4% (5.2, 36.3)	26.4% (12.1, 51.8)	45.3% (23.8, 73.7)
“	Highest risk (diabetes, high BP, & meds)	12.7% (2.3, 54.3)	23.4% (5.0, 74.7)	40.8% (10.1, 92.5)
Women	*Baseline (no risk)	0.3% (0.1, 0.8)	1.1% (0.6, 2.0)	4.3% (2.4, 7.5)
“	*Diabetes only	0.5% (0.1, 2.4)	2.1% (0.7, 6.8)	8.2% (2.7, 23.7)
“	*High BP only	0.7% (0.2, 2.3)	2.6% (1.2, 5.6)	10.1% (5.5, 18)
“	*Diabetes & High BP	1.3% (0.3, 6.2)	5.2% (1.5, 16.5)	18.9% (6.5, 47.9)
“	**Highest risk (+ Anti-Hyp. Meds)	11.1% (2.5, 42.4)	16% (4.9, 45.4)	22.8% (7.8, 56.1)

*Risk profile is for a patient in the 10th percentile for serum total cholesterol

**Risk profile is for a patient in the 90th percentile for serum total cholesterol

Table 4: Transition probabilities between baseline (period 1) and period 2, and between period 2 and period 3

Risk Factor	Periods	Men (n=1359)			Women (n=1838)		
*Diabetes Status	Period 1 to Period 2	Status at Period 1	Status at Period 2		Status at Period 1	Status at Period 2	
			No	Yes		No	Yes
		No	0.98	0.02	No	0.99	0.01
		Yes	-	1.00	Yes	-	1.00
	Period 2 to Period 3	Status at Period 2	Status at Period 3		Status at Period 2	Status at Period 3	
			No	Yes		No	Yes
		No	0.95	0.05	No	0.96	0.04
		Yes	-	1.00	Yes	-	1.00
**Systolic BP status	Period 1 to Period 2	Status at Period 1	Status at Period 2		Status at Period 1	Status at Period 2	
			No	Yes		No	Yes
		No	0.93	0.07	No	0.91	0.09
		Yes	0.39	0.61	Yes	0.30	0.70
	Period 2 to Period 3	Status at Period 2	Status at Period 3		Status at Period 2	Status at Period 3	
			No	Yes		No	Yes
		No	0.87	0.13	No	0.86	0.14
		Yes	0.43	0.57	Yes	0.35	0.65

*Prevalence of diabetics among men and women by period 3 is 8.7% and 7.2% respectively

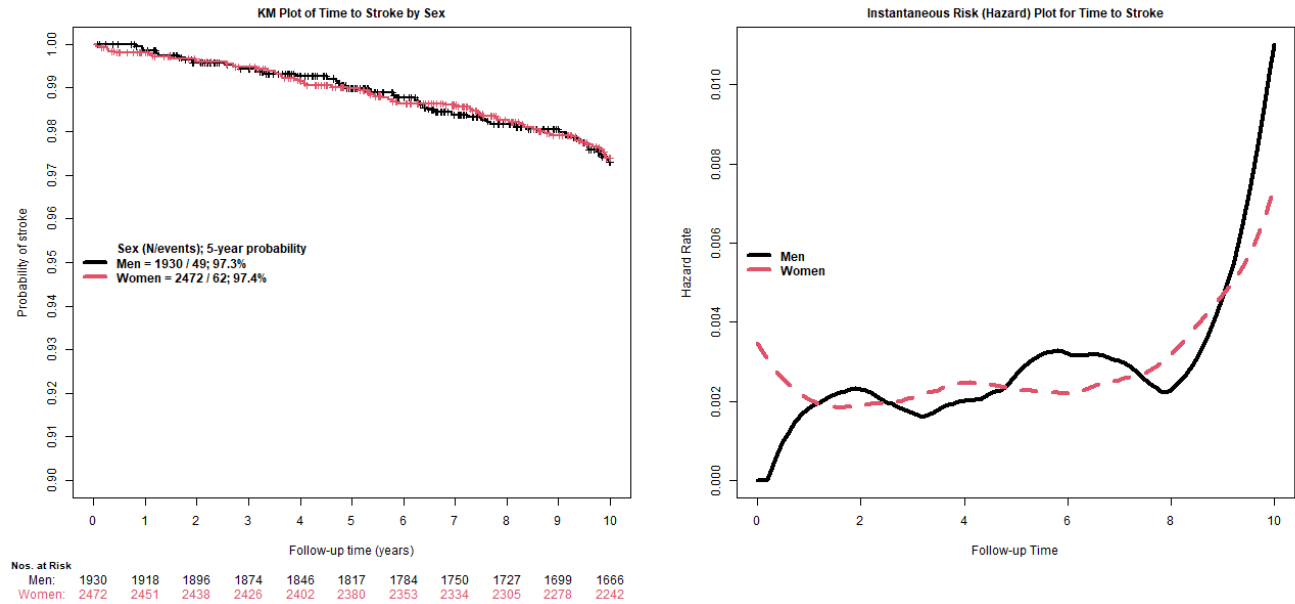
** Prevalence of high BP among men and women by period 3 is 17.3% and 22.0% respectively

Name: Opeyemi Jegede

Date: 04/17/2026

Project 3

Figure 1: Plot of Kaplan-Meier (KM) estimate of time to stroke [left panel] and baseline hazard plot for stroke [right panel] by sex

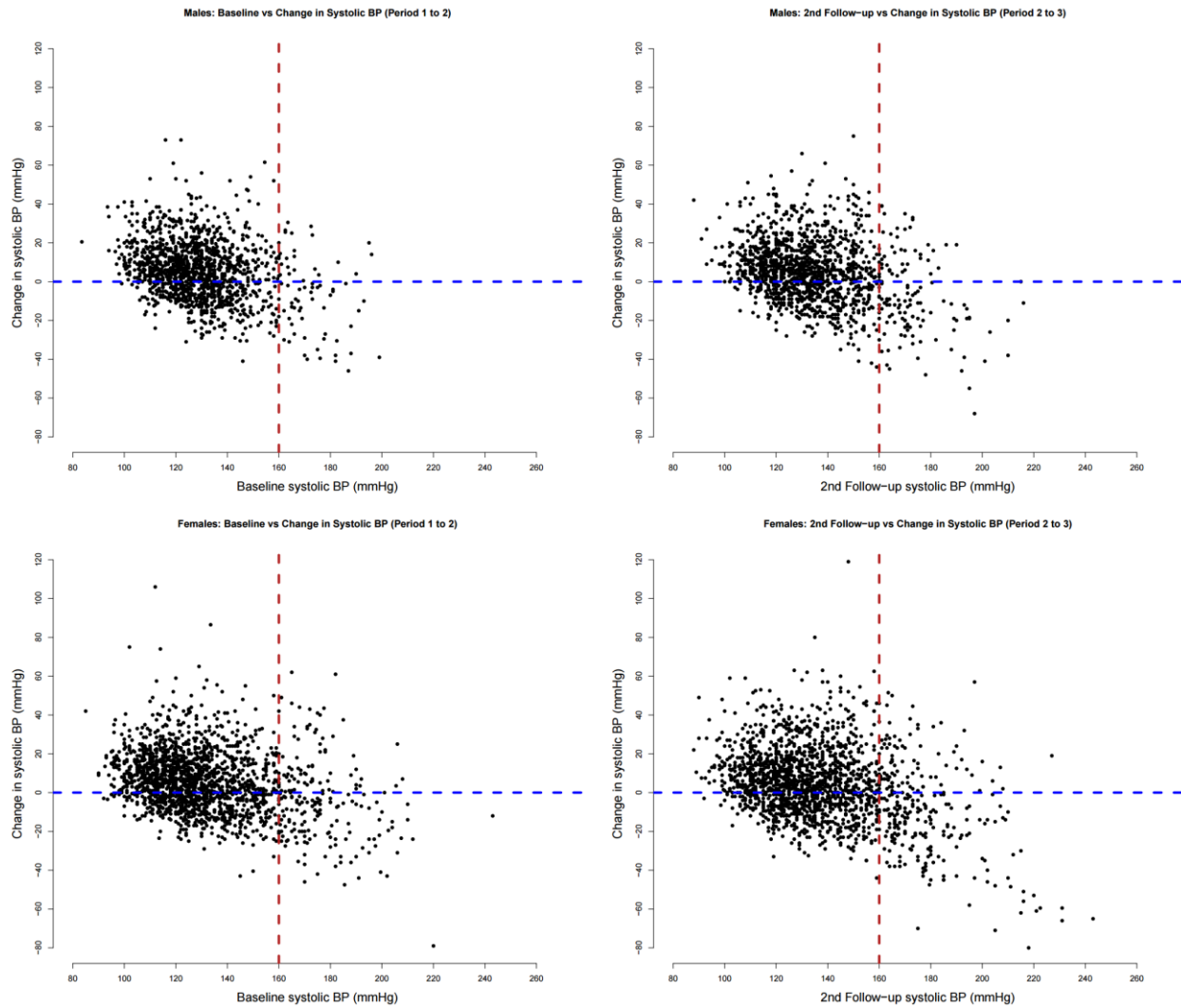


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Project 3

Figure 2: Scatter plot of change in systolic blood pressure (BP) against reference systolic BP measurements; dashed blue line represents no change between 2 timepoints, and the red line shows the ≥ 160 mmHg cutoff threshold.



Name: Opeyemi Jegede

Date: 04/17/2026

Project 3

References

1. Terry Therneau. 1996. Extending the Cox Model. Technical report series, Mayo Clinic, Rochester, Minnesota. Section 4, page 16. Assessed on 04/18/2026.
<https://www.mayo.edu/research/documents/biostat-58pdf/doc-10027288>

Reproducible Research Information

- GitHub repository for Project 3: <https://github.com/yemiaj/BIOS6624/tree/main/Project3>
- Raw data file (from Canvas) location:
“C:/Users/Owner/Documents/cuclasses/spring26/bios_6624/BIOS6624/Project3/DataRaw/frmgham2.csv”
- Processed and clean final analysis file location:
“C:\Users\Owner\Documents\cuclasses\spring26\bios_6624\BIOS6624\Project3\DataProcessed”
- Code files layout:
 - *“/Code/p3_clean_and_analyses.R”* is the main analysis program. Here you will find code to clean the raw data, save a clean version of the data as .csv in the /DataProcessed directory, and codes to create the tables, figures, supplementary tables and figures included in the final report.
 - *“/Code/ p3_thoughts.R”* contains thoughts on analysis plan, notes taken in class during project discussion, and chunk of rough/preliminary analysis codes. This is not the main analysis program.
- Main report
 - *“C:\Users\Owner\Documents\cuclasses\spring26\bios_6624\BIOS6624\Project3\Reports/Jegede_Project3_FraminghamStudy_RiskFactors_and_StrokeProbability.pdf”* is the final study report.
- README files in the main and sub-directories were updated with comments to make them more informative.

Supplementary Results**Supplementary Table 1: Participants characteristics at baseline by sex and stroke event status**

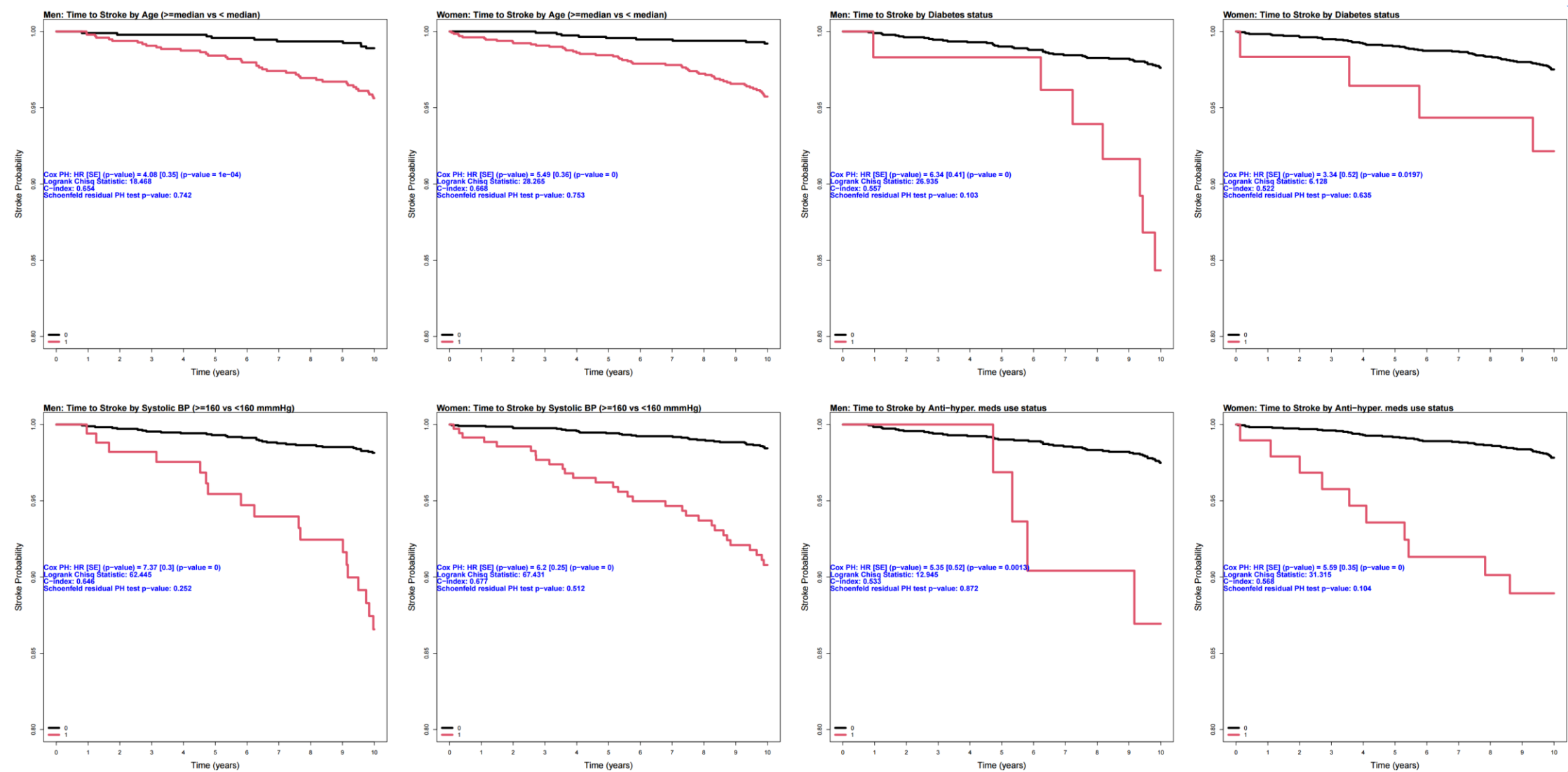
Participants' Baseline Characteristics	Participants' Sex and Stroke Status Combination			
	Men/No Stroke N = 1,881 ¹	Men/Yes Stroke N = 49 ¹	Women/No Stroke N = 2,410 ¹	Women/Yes Stroke N = 62 ¹
Age at exam (years)	49.6 (8.7), [49.0]	55.8 (7.5), [57.0]	49.8 (8.6), [49.0]	57.1 (7.3), [59.0]
Diabetic (casual glucose $\geq$ 200 mg/dL)				
Diabetic	52 (3%)	7 (14%)	56 (2%)	4 (6%)
Not Diabetic	1,829 (97%)	42 (86%)	2,354 (98%)	58 (94%)
Average systolic blood pressure (mmHg)	131.1 (18.9), [128.0]	150.9 (27.6), [143.0]	132.9 (23.6), [128.0]	163.6 (33.6), [158.5]
Systolic blood pressure status				
High Systolic BP ($\geq$ 160 mmHg)	153 (8%)	18 (37%)	323 (13%)	30 (48%)
Normal Systolic BP (<160 mmHg)	1,728 (92%)	31 (63%)	2,087 (87%)	32 (52%)
Use of anti-hypertensive medication				
Current Use	36 (2%)	4 (8%)	86 (4%)	10 (17%)
Not Currently Used	1,824 (98%)	44 (92%)	2,289 (96%)	49 (83%)
NA (missing)	21 (1.1%)	1 (2.0%)	35 (1.5%)	3 (4.8%)
Prevalent Coronary Heart Disease				
Prevalent Disease	115 (6%)	5 (10%)	60 (2%)	7 (11%)
Free of Disease	1,766 (94%)	44 (90%)	2,350 (98%)	55 (89%)
Current cigarette smoking				
Current Smoker	1,136 (60%)	32 (65%)	980 (41%)	21 (34%)
Not Current Smoker	745 (40%)	17 (35%)	1,430 (59%)	41 (66%)
Serum total cholesterol (mg/dL)	233.5 (42.3), [231.0]	238.6 (45.9), [229.0]	239.3 (46.2), [237.0]	253.6 (45.6), [253.5]
NA (missing)	7 (0.4%)	0 (0%)	43 (1.8%)	2 (3.2%)
Body Mass Index (kg/m ²)	26.2 (3.4), [26.1]	26.6 (4.1), [26.8]	25.5 (4.5), [24.8]	27.3 (5.5), [27.2]
NA (missing)	4 (0.2%)	0 (0%)	13 (0.5%)	0 (0%)
BMI Categories				
Underweight	12 (1%)	0 (0%)	44 (2%)	1 (2%)
Healthy Weight	680 (36%)	17 (35%)	1,204 (50%)	22 (35%)
Overweight	962 (51%)	25 (51%)	823 (34%)	24 (39%)
Obese	223 (12%)	7 (14%)	326 (14%)	15 (24%)
NA (missing)	4 (0.2%)	0 (0%)	13 (0.5%)	0 (0%)
Stroke event status				
Stroke Event	0 (0%)	49 (100%)	0 (0%)	62 (100%)
No Stroke Event	1,881 (100%)	0 (0%)	2,410 (100%)	0 (0%)
Nos. missing values for key variables				
No missing data	1,849 (98%)	48 (98%)	2,321 (96%)	57 (92%)
Missing for 1 variable	32 (2%)	1 (2%)	87 (4%)	5 (8%)
Missing for 2 variables	0 (0%)	0 (0%)	2 (0%)	0 (0%)

¹Mean (SD), [Median]; n (%)

Supplementary Table 2: Pairwise assessment of interaction between predictors for men and women

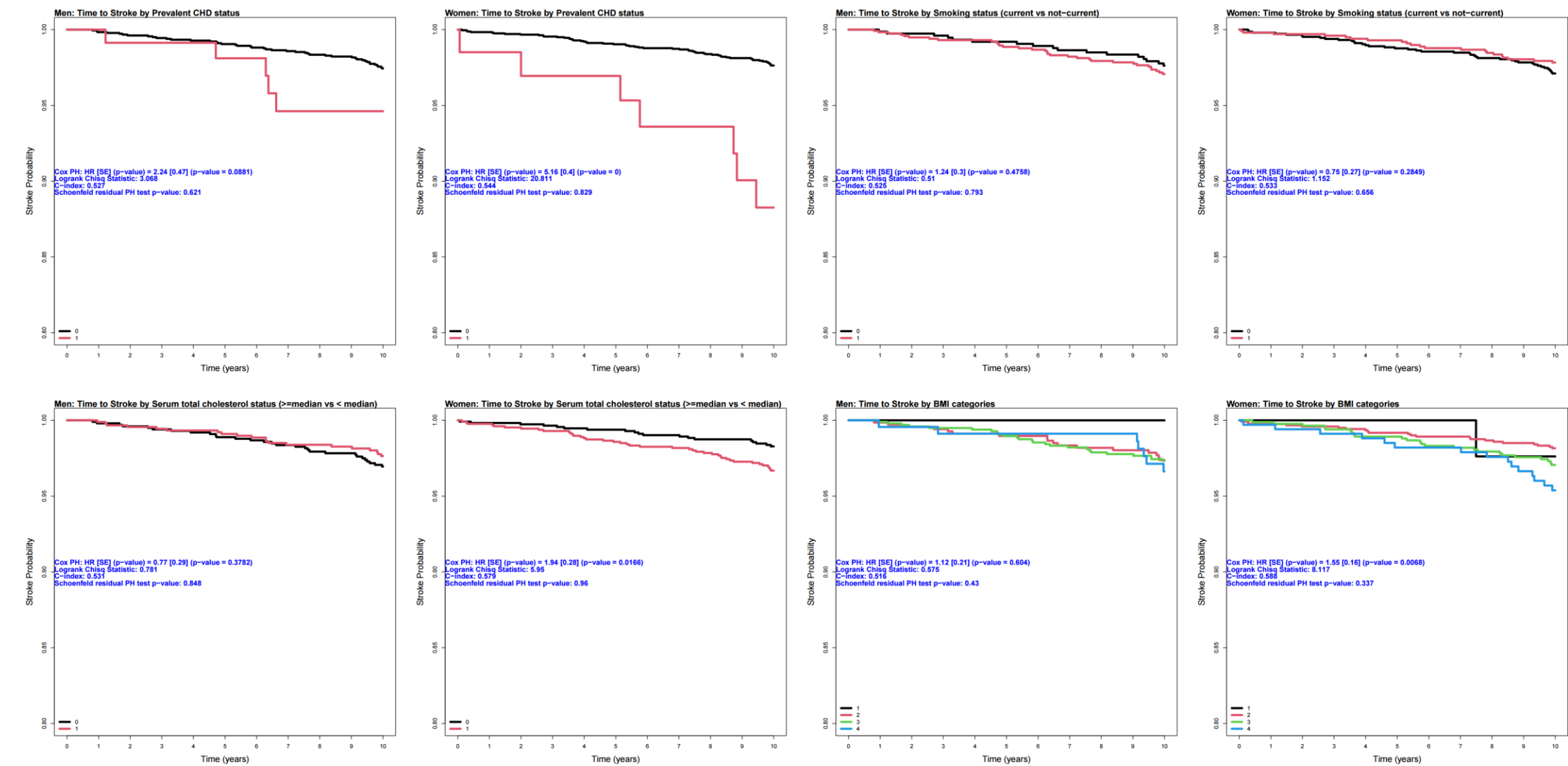
Sex	Model	neg2LogLik	LRTstat	LRTstat.pvalue
Men	Baseline (AGE + DIABETES + SYSBP.high + CURSMOKE)	653.752	0.000	1.000
	Main effects (+ BPMEDS)	652.207	1.545	0.214
	AGE:BPMEDS int.	651.718	2.033	0.154
	DIABETES:BPMEDS int.	651.900	1.852	0.174
	SYSBP.high:BPMEDS int.	647.627	6.125	0.013
	CURSMOKE:BPMEDS int.	652.085	1.667	0.197
	Baseline (AGE + DIABETES + SYSBP.high + CURSMOKE)	653.752	0.000	1.000
	Main effects (+ PREVCHD)	653.752	0.000	0.996
	AGE:PREVCHD int.	653.038	0.714	0.398
	DIABETES:PREVCHD int.	651.833	1.919	0.166
	SYSBP.high:PREVCHD int.	652.971	0.781	0.377
	CURSMOKE:PREVCHD int.	653.749	0.003	0.956
	Baseline (AGE + DIABETES + SYSBP.high + CURSMOKE)	653.752	0.000	1.000
	Main effects (+ TOTCHOL)	653.183	0.569	0.451
	AGE:TOTCHOL int.	652.717	1.034	0.309
	DIABETES:TOTCHOL int.	653.142	0.609	0.435
	SYSBP.high:TOTCHOL int.	653.165	0.587	0.444
	CURSMOKE:TOTCHOL int.	652.722	1.030	0.310
	Baseline (AGE + DIABETES + SYSBP.high + CURSMOKE)	653.752	0.000	1.000
	Main effects (+ BMI)	653.579	0.172	0.678
	AGE:BMI int.	652.605	1.146	0.284
	DIABETES:BMI int.	653.579	0.173	0.678
	SYSBP.high:BMI int.	653.570	0.182	0.670
	CURSMOKE:BMI int.	653.577	0.175	0.676
Women	Baseline (AGE + DIABETES + SYSBP.high + BPMEDS)	818.360	0.000	1.000
	Main effects (+ PREVCHD)	817.095	1.264	0.261
	AGE:PREVCHD int.	814.628	3.732	0.053
	DIABETES:PREVCHD int.	816.254	2.105	0.147
	SYSBP.high:PREVCHD int.	816.710	1.650	0.199
	BPMEDS:PREVCHD int.	816.147	2.212	0.137
	Baseline (AGE + DIABETES + SYSBP.high + BPMEDS)	818.360	0.000	1.000
	Main effects (+ CURSMOKE)	817.254	1.106	0.293
	AGE:CURSMOKE int.	815.500	2.859	0.091
	DIABETES:CURSMOKE int.	817.245	1.114	0.291
	SYSBP.high:CURSMOKE int.	816.512	1.848	0.174
	BPMEDS:CURSMOKE int.	814.237	4.122	0.042
	Baseline (AGE + DIABETES + SYSBP.high + BPMEDS)	818.360	0.000	1.000
	Main effects (+ TOTCHOL)	818.134	0.226	0.635
	AGE:TOTCHOL int.	813.675	4.684	0.030
	DIABETES:TOTCHOL int.	817.941	0.418	0.518
	SYSBP.high:TOTCHOL int.	817.628	0.732	0.392
	BPMEDS:TOTCHOL int.	816.741	1.619	0.203
	Baseline (AGE + DIABETES + SYSBP.high + BPMEDS)	818.360	0.000	1.000
	Main effects (+ BMI)	818.268	0.091	0.763
	AGE:BMI int.	817.753	0.606	0.436
	DIABETES:BMI int.	817.686	0.674	0.412
	SYSBP.high:BMI int.	817.943	0.416	0.519
	BPMEDS:BMI int.	818.118	0.242	0.623

Supplementary Figure 1a: Univariable analysis results for predictors of men and women; red line is coded to represent higher risk



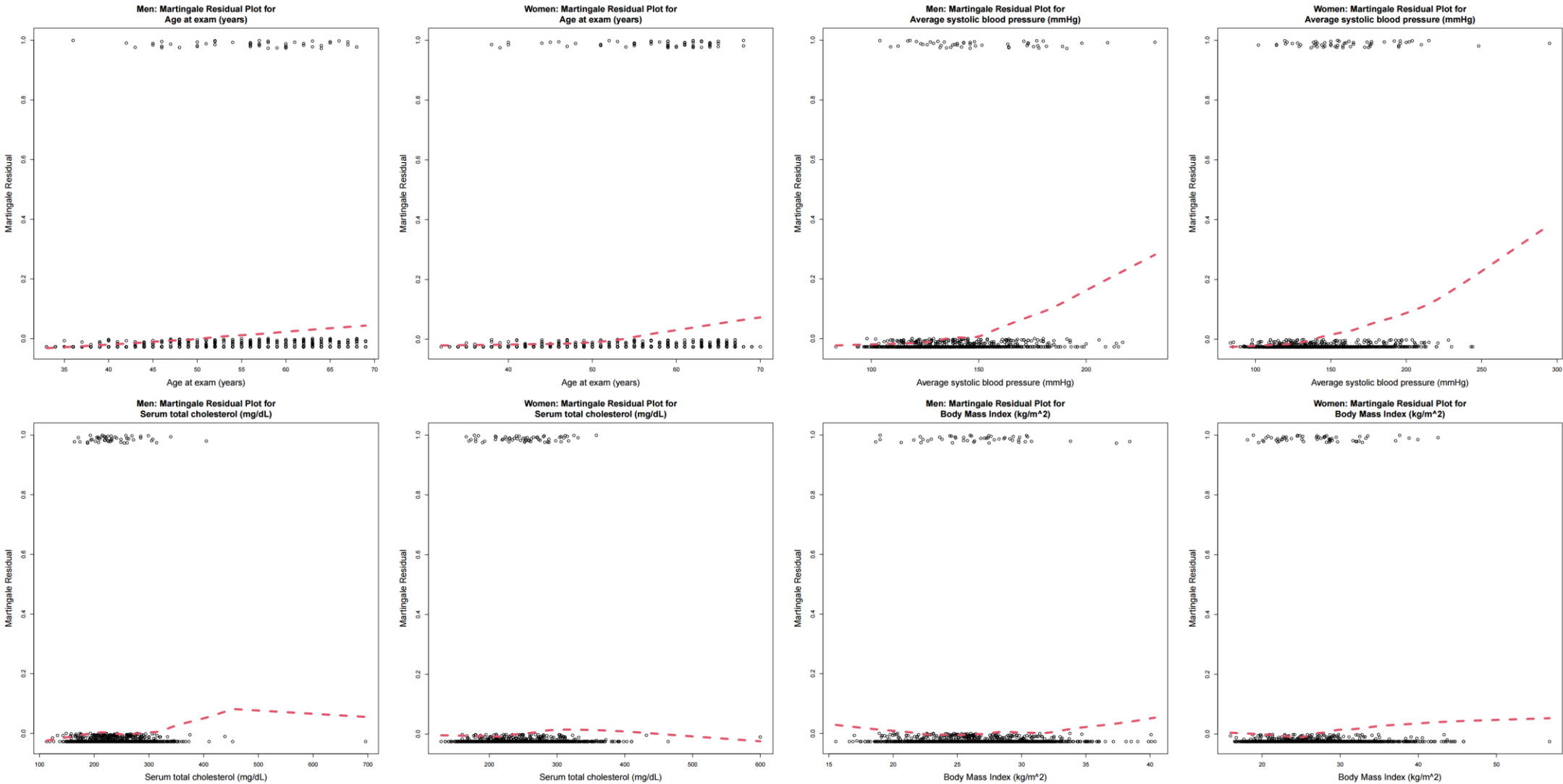
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Supplementary Figure 1b: (continued) Univariable analysis results for predictors or men and women; red line is coded to represent higher risk



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Supplementary Figure 2: Martingale residual analysis for continuous variables, for men and women



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